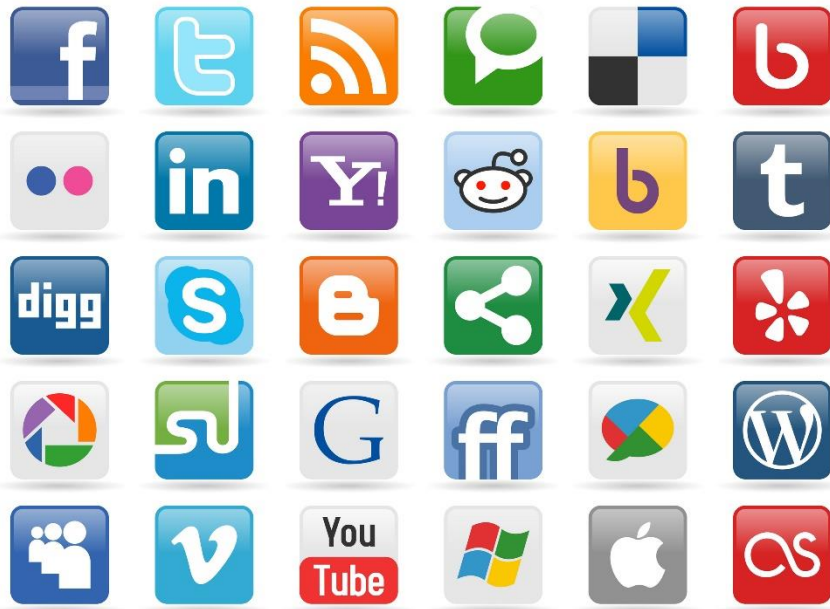




Recommendation in Social Media

SOCIAL MEDIA MINING



When Does This Problem Occur?

- There are many choices
- There are no obvious advantages among them
- We do not have enough resources to check all options (**information overload**)
- We do not have enough knowledge and experience to choose, or
 - I'm lazy, but don't want to miss out on good stuff

Goal of Recommendation:
To come up with a short list of items that fits user's interests

Common Solutions to the Problem

- Consulting friends
- Obtaining information from a trusted third party
- Hiring a team of experts
- Search the Internet
- Following the crowd
 - Pick the item from top- n lists
 - Best sellers on Amazon
- **Can we automate all of the above?**
 - Using a recommender algorithm
 - Also known as **recommender systems**

Main Idea behind Recommender Systems

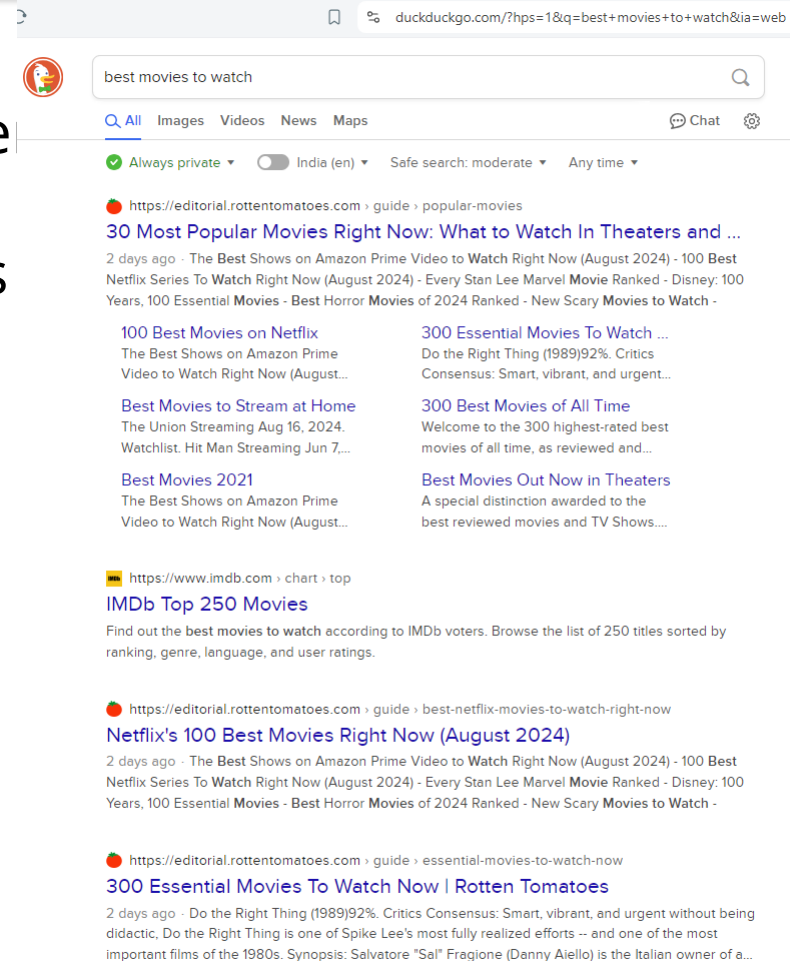
Use historical data such as the user's past preferences or similar users' past preferences to predict future likes

- Users' preferences are likely to remain stable, and change smoothly over time.
 - By watching the past users' or groups' preferences, we try to predict their future interests
 - Then we can recommend items of interest to them
- Formally, a recommender system takes a set of users U and a set of items I and learns a function f such that:

$$f : U \times I \rightarrow \mathbb{R}$$

Recommendation vs. Search

- **Search engines** find results that match the query provided by the user
- The results are generally provided as a list ordered with respect to the relevance of the item to the given query
- Consider the query “**best movies to watch**”
 - Results for different users



Search engines' results need not be customized

Challenges of Recommender Systems

- **The Cold Start Problem**

- Recommender systems use historical data or information provided by the user to recommend items, products, etc.
- When user join sites, they still haven't bought any product, or they have no history.
- It is hard to infer what they are going to like when they start on a site.

- **Data Sparsity**

- When historical or prior information is insufficient.
- Unlike the cold start problem, this is in the system as a whole and is not specific to an individual.

Challenges of Recommender Systems

- **Attacks**

- Push ratings up by making fake users

- **Privacy**

- Using one's private info to recommend to others.

- **Explanation**

- Recommender systems often recommend items with no explanation on why these items are recommended

Explainable Recommender Systems



Fig. 1. Trustworthiness is a multi-perspective concept in AI and recommender systems.

<https://doi.org/10.1145/3652891>

Classical Recommendation Algorithms

- **Content-based algorithms**
- **Collaborative filtering**

Content-Based Methods

Assumption: a user's interest should match the description of the items that the user should be recommended by the system.

- The more similar the item's description to that of the user's interest, the more likely that the user finds the item's recommendation interesting.

Goal: finding the similarity between the user and all the existing items

Content-based Recommendation: An Example

Book Database

Title	Genre	Author	Type	Price	Keywords
<i>The Night of the Gun</i>	Memoir	David Carr	Paperback	29.90	press and journalism, drug addiction, personal memoirs, New York
<i>The Lace Reader</i>	Fiction, Mystery	Brunonia Barry	Hardcover	49.90	American contemporary fiction, detective, historical
<i>Into the Fire</i>	Romance, Suspense	Suzanne Brockmann	Hardcover	45.90	American fiction, murder, neo-Nazism
...					

User Profile

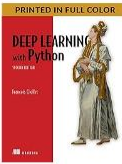
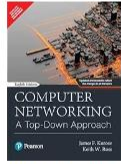
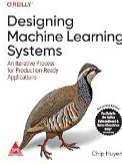
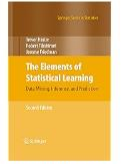
Title	Genre	Author	Type	Price	Keywords
...	Fiction, Suspense	Brunonia Barry, Ken Follett	Paperback	25.65	detective, murder, New York

Content-based Recommendation Algorithm

1. Describe the items to be recommended
2. Create a profile of the user that describes the types of items the user likes
3. Compare items with the user profile to determine what to recommend

The profile is often created, and updated automatically in response to feedback on the desirability of items that are presented to the user

Content-based Recommendation: Example

 Phonic Reader (Illustrated) (Set of 6 Books) - Story Book for Kids... ★★★★☆ 1,022 ₹163.00 Get it by Friday, September 6	 Hands-On Machine Learning with Scikit-Learn, Keras, and... ★★★★☆ 185 ₹3,250.00 Get it by Monday, September 9	 Deep Learning with Python, Second Edition ★★★★☆ 363 ₹4,940.00 Get it by Tuesday, September 10	 Wipro Smart Wireless Security CCTV Camera 3 MP 1296p Full... ★★★★☆ 233 ₹2,375.00 Get it by Tuesday, September 10
 Data Structures and Algorithms in Python, An Indian Adaptation ★★★★☆ 263 ₹662.00 Get it by Monday, September 9	 Computer Networking, 8e ★★★★☆ 176 ₹819.00 Get it by Friday, September 6	 Designing Machine Learning Systems: An Iterative Process for... ★★★★☆ 594 ₹1,600.00 Get it by Friday, September 6	 imusti The Elements Of Statistical Learning: Data Mining, Inference, And Prediction, Second Edition... ★★★★☆ 1,207 ₹1,470.00

User Profile

Items Recommended

1-16 of 118 results

Delivery Day

☐ Get it by Tomorrow

☐ Get It in 2 Days

Prime Reading

☐ Prime Reading Eligible

Category

< Books

< Literature & Fiction

Anthologies

Customer Reviews

★★★★☆ & Up

Price

₹68 – ₹650+

Go

Deals & Discounts

All Discounts

Today's Deals

Item Condition

New

Used

Kindle Unlimited

☐ Kindle Unlimited Eligible

English & Indian Languages

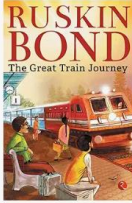
☐ English

Results

Check each product page for other buying options.



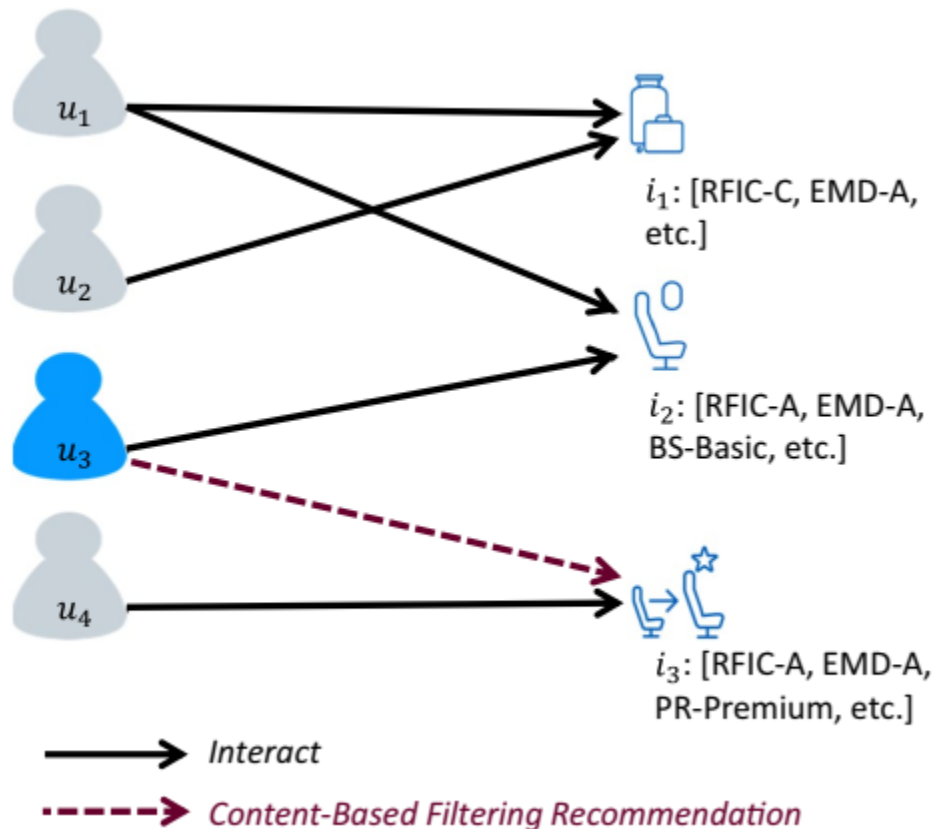
THE GIRL ON THE TRAIN
by Ruskin Bond | 11 December 2023
★★★★☆ 126
Paperback
₹199 M.R.P: ₹250 (20% off)
Buy 3 items, get 3% off
More Buying Choices
₹175 (8 new offers)



The Great Train Journey
by Ruskin Bond | 20 August 2018
★★★★☆ 2,188
Paperback
₹129 M.R.P: ₹195 (34% off)
Buy 3 items, get 3% off
Audible Audiobook
₹0
Free with Audible trial
Available instantly
Audio CD
Currently unavailable.

FALLING IN LOVE AGAIN

Content-based illustration



<https://doi.org/10.1057/s41272-021-00313-2>

- item i_3 (premium seat) is recommended to user u_3 because item i_3 has the same characteristics of item i_2 which user u_3 has interacted with (added in his cart) in the past

More formally

- We represent user profiles and item descriptions by vectorizing them using a set of k keywords
- We can vectorize (e.g., using **TF-IDF**) both users and items and compute their similarity

$$I_j = (i_{j,1}, i_{j,2}, \dots, i_{j,k}) \quad U_i = (u_{i,1}, u_{i,2}, \dots, u_{i,k})$$

$$\text{sim}(U_i, I_j) = \cos(U_i, I_j) = \frac{\sum_{l=1}^k u_{i,l} i_{j,l}}{\sqrt{\sum_{l=1}^k u_{i,l}^2} \sqrt{\sum_{l=1}^k i_{j,l}^2}}$$

**We can recommend the top most
similar items to the user**

Text Representation

- The most common way to represent documents is to transform them into vectors
 - Process them with linear algebraic operations
- This representation is called “*Bag of Words*”
 - Vector Space Model
- Weights for words can be assigned by **TF-IDF**

Vector Space Model

- Consider a set of documents D
- Each document is a set of words
- **Goal:** convert these documents to vectors

$$d_i = (w_{1,i}, w_{2,i}, \dots, w_{N,i})$$

- d_i : document i
- $w_{j,i}$: the weight for word j in document i

How to set $w_{j,i}$

- Set $w_{j,i}$ to 1 when the word j exists in document i and 0 when it does not.
- We can also set $w_{j,i}$ to the number, of times the word j is observed in document i (frequency)

Vector Space Model: An Example

- **Documents:**
 - d_1 : social media mining
 - d_2 : social media data
 - d_3 : financial market data
- Reference vector (**Dictionary**):
 - (social, media, mining, data, financial, market)
- Vector representation:

	social	media	mining	data	financial	market
d_1	1	1	1	0	0	0
d_2	1	1	0	1	0	0
d_3	0	0	0	1	1	1

TF-IDF (Term Frequency-Inverse Document Frequency)

TF-IDF of term (word) t , document d , and document corpus D is calculated as follows:

$$w_{j,i} = tf_{j,i} \times idf_j$$

$tf_{j,i}$ is the frequency of word j in document i

$$idf_j = \log_2 \frac{|D|}{|\{\text{document} \in D \mid j \in \text{document}\}|}$$

The total number of documents in the corpus

The number of documents where the term j appears

TF-IDF: An Example

Document d_1 contains 100 words

- Word “apple” appears 10 times in d_1
- Word “orange” appears 20 times in d_1

We have $|D| = 20$ documents

- Word “apple” only appears in document d_1
- Word “orange” appears in all 20 documents

$$tf - idf(\text{“apple”}, d_1) = 10 \times \log_2 \frac{20}{1} = 43.22$$

$$tf - idf(\text{“orange”}, d_1) = 20 \times \log_2 \frac{20}{20} = 0$$

TF-IDF: An Example

- Documents:
 - d_1 : social media mining
 - d_2 : social media data
 - d_3 : financial market data

$$idf_{\text{social}} = \log_2(3/2) = 0.584$$

$$idf_{\text{media}} = \log_2(3/2) = 0.584$$

$$idf_{\text{mining}} = \log_2(3/1) = 1.584$$

$$idf_{\text{data}} = \log_2(3/2) = 0.584$$

$$idf_{\text{financial}} = \log_2(3/1) = 1.584$$

$$idf_{\text{market}} = \log_2(3/1) = 1.584$$

- TF values:

	social	media	mining	data	financial	market
d_1	1	1	1	0	0	0
d_2	1	1	0	1	0	0
d_3	0	0	0	1	1	1

- TF-IDF

	social	media	mining	data	financial	market
d_1	0.584	0.584	1.584	0	0	0
d_2	0.584	0.584	0	0.584	0	0
d_3	0	0	0	0.584	1.584	1.584

Content-Based Recommendation Algorithm

Algorithm 9.1 Content-based recommendation

Require: User i 's Profile Information, Item descriptions for items $j \in \{1, 2, \dots, n\}$, k keywords, r number of recommendations.

- 1: **return** r recommended items.
 - 2: $U_i = (u_1, u_2, \dots, u_k)$ = user i 's profile vector;
 - 3: $\{I_j\}_{j=1}^n = \{(i_{j,1}, i_{j,2}, \dots, i_{j,k}) = \text{item } j\text{'s description vector}\}_{j=1}^n$;
 - 4: $s_{i,j} = \text{sim}(U_i, I_j)$, $1 \leq j \leq n$;
 - 5: Return top r items with maximum similarity $s_{i,j}$.
-

- We compute the topmost similar items to a user j and then recommend these items in the order of similarity

(Dis)advantages of Content-based Recommendation

- Does not leverage information from another similar user
- Lack of novelty
- Useful for cold-start problems